



CMP3050

One Time Pad Stream cipher

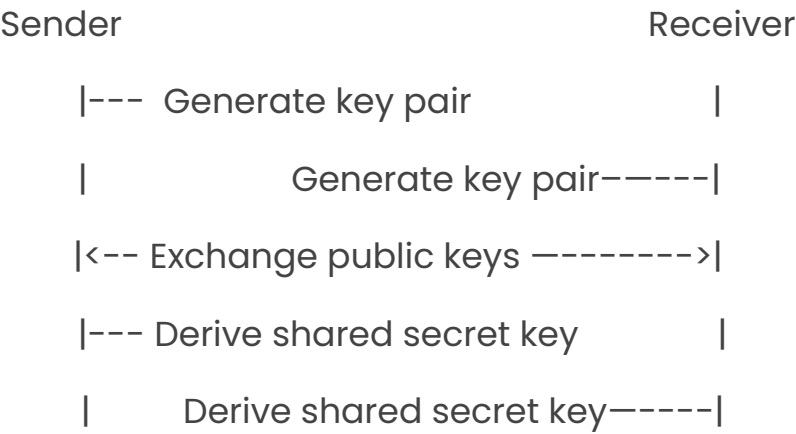
Team 22

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Shehab Khaled	9220393	Sender - LCG - SocketSender - Stream Cipher - Key Exchange

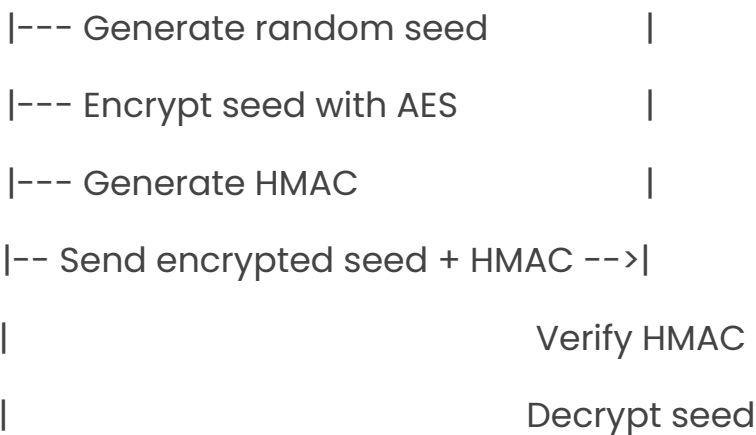
1. System Architecture

The project implements a secure communication system between two parties (sender and receiver) using various cryptographic primitives. The system is built with a modular design, separating concerns into distinct components.

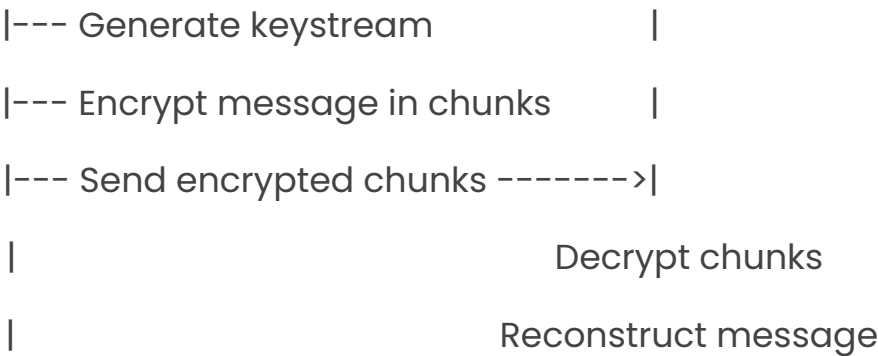
setup-----



Seed Exchange-----



Message Exchange-----



2. Used Algorithms:

2.1. Key Exchange with Diffie-Hellman:

Diffie-Hellman (DH) key exchange relies on predefined, cryptographically secure groups to ensure safety against attacks. The RFC 3526 standard defines several such groups, which are widely used in TLS, VPNs, and other secure protocols.

used group: Group 14 (2048-bit)

- $p = 2^{2048} - 2^{1984} - 1 + 2^{64} * ((2^{1918} \pi) + 124476)$
- $\alpha = 2$

2.2. Message Authentication with HMAC-SHA256:

- SHA-256 is widely supported and considered secure.
- The output is 32 bytes (256 bits), offering a good balance between security and performance
- for the same key and the same message, generated HMAC will always be the same.

2.3. Linear Congruential Generator (LCG):

A widely used technique for pseudorandom number generation, in our case used for generating the one-time pad key given a seed for plaintext encryption

$$X_{n+1} = (aX_n + c) \bmod (m)$$

- Required characteristics for LCG:
 1. The function should be a full-period generating function. That is, the function should generate all the numbers between 0 and m before repeating.
 2. The generated sequence should appear random.
 3. The function should be implemented efficiently with 32-bit arithmetic.
- With appropriate values of a, m and c, these three tests can be passed. With respect to m, it can be shown that if m is prime and $c = 0$, then for

certain values of a the period of the generating function is $m - 1$, with only the value 0 missing.

- for 32-bit arithmetic, a convenient prime value of is $2^{31} - 1$.
- Of the more than 2 billion possible choices for a , only a handful of multipliers

pass all three tests. One such value is 7^5 , which was originally selected for use in the IBM 360 family of computers [LEWI69].

thus the parameters chosen are:

- $a = 7^5, m = 2^{31} - 1, c = 0$

2.4. Seed Encryption with AES-256:

A symmetric block cipher (powerful alternative to DES), in our case used to encrypt the seed.

- $\text{key_size} = 32 \text{ bytes} = 256 \text{ bits}$ which is the strongest variant of AES among the three standard options (AES-128, AES-192, AES-256)
- creates a random 16-byte initialization vector (IV) which is crucial for CBC mode to ensure identical plaintexts produce different ciphertexts
- plaintext size = 128 bits which is the randomly generated seed in our case

2.5. Stream Cipher with OTP:

uses a random key that is as long as the message (generated with LCG), so that the key need not be repeated. In addition, the key is to be used to encrypt and decrypt a single message, and then is discarded. Each new message requires a new key of the same length as the new message. Such a scheme, known as a **one-time pad**, is **unbreakable**. It produces random output that bears **no statistical relationship** to the plaintext. Because the ciphertext contains no information whatsoever about the plaintext, there is simply no way to break the code.

3. Testing:

Test 1:

Plain Text:

be aware!
you're in danger

Sender Side:

Message to send: b"be aware! \nyou're in danger"

Random seed: 316524314335238904560423980819765496118

Sender Public Key Components:

Prime modulus (p):

0xffffffffffffffc90fdaa22168c234c4c6628b80dc1cd129024e088a67cc74020bbe
a63b139b22514a08798e3404ddef9519b3cd3a431b302b0a6df25f14374fe1356d6
d51c245e485b576625e7ec6f44c42e9a637ed6b0bff5cb6f406b7edee386bfb5a
899fa5ae9f24117c4b1fe649286651ece45b3dc2007cb8a163bf0598da48361c55d
39a69163fa8fd24cf5f83655d23dca3ad961c62f356208552bb9ed529077096966
d670c354e4abc9804f1746c08ca18217c32905e462e36ce3be39e772c180e86039
b2783a2ec07a28fb5c55df06f4c52c9de2bcbf6955817183995497cea956ae515d
2261898fa051015728e5a8aaca68ffffffffffffff

Generator (g): 0x2

Sender public key:

0x92404bda0646044f6df265926de088c30e4a856a85988ff30929c3708eb8164
7278c2e172ada25d86e63931c383d5a15599fac59dde9ffe76147e9d2afd5db847
a6e65d828cd93eb10faf3c8914897a3d2f189097fd2c9947103757c258fe47a87641
3fb9d97b5b48ab8b02ee508d9be3ef980c0c739c7f2515b3479fd83669774ac85
b8e6b9b69ff86808268a4baeaf23a2dae077612c409f8163dbc0ce52f0f0ea7b09
3e3e160f72a2c52d796404aa535f439b227f4ce4715d85bbeee03f218317e9d557e
cd0615e0cd534293a5cc97a0dabe00a26d4e7195fb157f41f32e2161a0f8d5ddale
550flabbf6669d2bb72f2f25dff7ac179268387c8bf290be9

Connected to receiver

INSIDE KEY EXCHANGE (peer public key):

0x4f602c1bcbab1a540f2eb02deec493a5e315c350f6a51ead1e4bcb9ac337fee27
c36b269734295326c713ca8d4f8f4317d04212b5e8a6070ad69e8f1ff2c4ad22b11a
563c6fc7a08e3bc518d01794a6f6ae64e0715e5fa40a22a2a60012b351f357e4bb6
d696f318acc61b5aa4b4ec983fffce4c630d75a04c5fa9c0ade5ec0621d0481308
54b16132c26fa4da8b51277323f895fb4b7788f95b9ab3b35e84579d5f8cff5a0a4
b08dcaed58974fbfa89109fe698446a7aad24d0fca277bcd74e1dc5d9a73e3100
b51b03fbca102d91864602188f7c03d041e9fa5d9500fcb65e7172386c326661720e
660eceed722072dba802a62dcb4ddfe0a8a593239a957

Shared key length: 256 bytes

Shared key:

b'\xa3\xff\xab\x19\xd1\x00l\xdd\xd0\x06wyQ\xc0\xa2\xd7P\xca\xb6""\xfb
\x1d+\t\x83\xf2\r\xf0>nc.F\x86\x87\x174l]\x02\xd88(@\x06\xbar\xfc\xe5U
\xad\xf7F\x9b\x05\xc2\x80\xc4\xc34\t\x8d\xba\xc6\xa2\x04-\xfe\xad\
x8e\xabS\xa2\x00C\xdc\x81hY\xd2\tK\xb57\x1d\xe6\xba\xaa\xda\xe7\x8
1\xb5\xf9\x93\x0eQw\x97\xdb\xf2K\x13Q\x1a\xb52Sa\xc8\xd5\xa3\xd3\x1
f2\x91\x12'\xa90\xe0\x1a?\x06\xd3V\xf9\xa0\xd9b#"'\xd3\x94\xed\xadZ\x
a1\x11z\xe9\r]iN\x98\x16\x9d\x11\xcf\xbc\x90v\xe5\x07\xc4o,-\x05}\xees{
\x9b\xaf\xca2\xb1\xf0\xbf\xdf\x8d\x0e\x1b\xe6\xfb\xce9;fo=\xc0\x16\xa
e)c\x1d\xcekb<W\x9fwnVL\xbfO"s\xc5hX#\xe3wT\x8f]\x1d\x035\xee\rz)\xf
4};\xe9\xf0R\x04\x99\x9fE\x16\xbd\nG\xbf\xf5\xe9m\xd0\xd2\xc3wq\xfa
\xe5\x11JK\xb6\xe2\xe0\x02\x02h\xde{\xb7'

INSIDE SEED ENCRYPTOR (key):

b'\r\x0cG\x84=\xb0\xf2\x11Q<BX5,@w\xe6tV\x1c\xc5\x9a\xb6\xc4\xee\xfb
6\xb5\x97\xca\xaf\xfc1' of length 32 bytes

INSIDE SEED ENCRYPTOR (seed): 316524314335238904560423980819765496118

Encrypted seed:

b'C\x8e\xb2%\xb3=\xba\xb3\xe2\xa91\xef.NK#dL1\x0c\x01\xb7\x1aA\x8a\
\x17!\x81j3\x87h\xc93\x82Y@\x11om\xdc\xc9\x89\xb8\xf9B\$X'

Encrypted seed length: 48 bytes

INSIDE HMAC:

b'snY\xclu\x9cpp\x9a\xben%\xdaU!\x80\xca&n\x19\ \x\x9aXzm7\xe7\x88
O' of length 32 bytes

INSIDE STREAM CIPHER: b"be aware!\nyou're in danger"

Cipher text:

b'\xc3\xa3\xca\x00\x05\ '\xf8\xe8\x8a\xe8"\xea\x96\$^%6\xe2\xe6\xa9\
xe7\t\xcfk\xdd' of length 26 bytes

Sent chunk 1/3

Sent chunk 2/3

Sent chunk 3/3

All chunks sent

Receiver Side:

Receiver Public Key Components:

Prime modulus (p):

0xffffffffffffc90fdaa22168c234c4c6628b80dc1cd129024e088a67cc74020bbe
a63b139b22514a08798e3404ddef9519b3cd3a431b302b0a6df25f14374fe1356d6
d51c245e485b576625e7ec6f44c42e9a637ed6b0bff5cb6f406b7edee386bf5a
899fa5ae9f24117c4b1fe649286651ece45b3dc2007cb8a163bf0598da48361c55d
39a69163fa8fd24cf5f83655d23dca3ad961c62f356208552bb9ed529077096966
d670c354e4abc9804f1746c08ca18217c32905e462e36ce3be39e772c180e86039
b2783a2ec07a28fb5c55df06f4c52c9de2bcbf6955817183995497cea956ae515d
2261898fa051015728e5a8aaca68ffffffffffff

Length of prime modulus (p): 256 bytes

Generator (g): 0x2

Receiver public key:

0x4f602c1bcbab1a540f2eb02deec493a5e315c350f6a51ead1e4bcb9ac337fee27
c36b269734295326c713ca8d4f8f4317d04212b5e8a6070ad69e8f1ff2c4ad22b11a
563c6fc7a08e3bc518d01794a6f6ae64e0715e5fa40a22a2a60012b351f357e4bb6
d696f318acc61b5aa4b4ec983fffce4c630d75a04c5fa9c0ade5ec0621d0481308

54b16132c26fa4da8b51277323f895fb4b7788f95b9ab3b35e84579d5f8cff5a0a4
b08dcaed58974fbfa89109fe698446a7aad24d0fca277bcd74e1dc5d9a73e3100
b51b03fbca102d91864602188f7c03d041e9fa5d9500fcb65e7172386c326661720e
660eceed722072dba802a62dcb4ddfe0a8a593239a957

Waiting for connection...

Connected by ('127.0.0.1', 56542)

INSIDE KEY EXCHANGE (peer public key):

0x92404bda0646044f6df265926de088c30e4a856a85988ff30929c3708eb8164
7278c2e172ada25d86e63931c383d5a15599fac59dde9ffe76147e9d2afd5db847a
6e65d828cd93eb10faf3c8914897a3d2f189097fd2c9947103757c258fe47a876413f
b9d97b5b48ab8b02ee508d9be3ef980c0c739c7f2515b3479fd83669774ac85b8
e6b9b69ff86808268a4baeaf23a2dae077612c409f8163dbc0ce52f0f0ea7b093e
3e160f72a2c52d796404aa535f439b227f4ce4715d85bbeee03f218317e9d557ecd
0615e0cd534293a5cc97a0dabe00a26d4e7195fb157f41f32e2161a0f8d5ddale550
flabbf6669d2bb72f2f25dff7ac179268387c8bf290be9

Shared key length: 256 bytes

Shared key:

b'\xa3\xff\xab\x19\xd1\x00l\xdd\xd0\x06wyQ\xc0\xa2\xd7P\xca\xb6"\xfb
\x1d+\t\x83\xf2\r\xf0>nc.F\x86\x87\x174l]\x02\xd88(@\x06\xbar\xfc\xe5U
\xad\xf7F\x9b\x05\xc2\x80\xc4\xc34\t\x8d\xba\xc6\xa2\x04-\xfe\xad\x
8e\xabS\xa2\x00C\xdc\x81hY\xd2\tK\xb57\x1d\xe6\xba\xaa\xda\xe7\x81
\xb5\xf9\x93\x0eQw\x97\xdb\xf2K\x13Q\x1a\xb52Sa\xc8\xd5\xa3\xd3\x1f
2\x91\x12'\xa90\xe0\x1a?\x06\xd3V\xf9\xa0\xd9b#\xd3\x94\xed\xadZ\xa
1\x11z\xe9\r]iN\x98\x16\x9d\x11\xcf\xbc\x90v\xe5\x07\xc4o,-\x05}\xees{\x
9b\xaf\xca2\xb1\xf0\xbf\xdf\x8d\x0e\x1b\xe6\xfb\xce9;fo=\xc0\x16\xae)c
\x1d\xcekb<W\x9fwnVL\xbfO"s\xc5hX#\xe3wT\x8f]\x1d\x035\xee\rz)\xf4};\n
xe9\xf0R\x04\x99\x9fE\x16\xbd\nG\xbf\xf5\xe9m\xd0\xd2\xc3wq\xfa\xe5
\x11JK\xb6\xe2\xe0\x02\x02h\xde{\xb7'

INSIDE HMAC:

b'snY\xclu\x9cpp\x9a\xben%"\xdaU!\x80\xca&n\x19\|\x\x9aXzm7\xe7\x88
O' of length 32 bytes

INSIDE SEED ENCRYPTOR (key):

b'\r\x0cG\x84=\xb0\xf2\x11Q<BX5,@w\xe6tV\x1c\xc5\x9a\xb6\xc4\xee\xf6
\xb5\x97\xca\xaf\xfc1' of length 32 bytes

Decrypted seed: 316524314335238904560423980819765496118

INSIDE STREAM CIPHER: b'{'\xc3\xa3\xca\x00\x05'\xf8\xe8\x8a"

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xe8"\xea\x96\$^%6\xe2\xe6'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xa9\xe7\t\xcfk\xdd'

Received chunk, decrypted 6 bytes

Decryption complete. Saved to decrypted.txt

Test 2:

Plain Text:

!@#\$%^&*()_+-=[{}|;:",".<>?/~`
ı™£¢∞§¶•ªº–≠œΣ∞†¥˜øπ™«≤÷
☀️👋🔒🔑💻📡🌐⚡🎯🎲
קטמנצח מרחבים עבריים والعربية

Sender Side:

Message to send:

b'!@#\$%^&*()_+-=[{}|;:",".<>?/~`n\xc2\xa1\xe2\x84\xa2\xc2\xa3\xc2\xa2
\xe2\x88\x9e\xc2\xa7\xc2\xb6\xe2\x80\xa2\xc2\xaa\xc2xba\xe2\x80\x
93\xe2\x89\xa0\xc5\x93\xe2\x88\x91\xc2\xb4\xc2xae\xe2\x80\xa0\xc2
\xa5\xc2\xa8\xcb\x86\xc3\xb8\xcf\x80"' \xc2\xab\xe2\x89\xa4\xe2\x89
\xa5\xc3\xb7\n\x0f\x9f\x8c\x9f \xf0\x9f\x91\x8b \xf0\x9f\x94\x92
\xf0\x9f\x97\x9d\xef\xb8\x8f \xf0\x9f\x92\xbb \xf0\x9f\x93\xa1
\xf0\x9f\x8c\x90 \xe2\x9a\xa1 \xf0\x9f\x8e\xaf
\xf0\x9f\x8e\xb2\n\xd7\x9e\xd7\xa8\xd7\x97\xd7\x91\xd7\x99\xd7\x9d
\xd7\xa2\xd7\x91\xd7\xa8\xd7\x99\xd7\x99\xd7\x9d
\xd9\x88\xd8\xa7\xd9\x84\xd8\xb9\xd8\xb1\xd8\xa8\xd9\x8a\xd8\xa9

\xe1\x80\xa1\xe1\x80\x80\xe1\x80\xb9\xe1\x80\x81\xe1\x80\x9b\xe1\x80\xac'

Random seed: 116197241997750167055934171051957910961

Sender Public Key Components:

Prime modulus (p):

0xffffffffffffc90fdaa22168c234c4c6628b80dc1cd129024e088a67cc74020bbea
63b139b22514a08798e3404ddef9519b3cd3a431b302b0a6df25f14374fe1356d6d
51c245e485b576625e7ec6f44c42e9a637ed6b0bff5cb6f406b7edee386bfb5a89
9fa5ae9f24117c4b1fe649286651ece45b3dc2007cb8a163bf0598da48361c55d39
a69163fa8fd24cf5f83655d23dca3ad961c62f356208552bb9ed529077096966d6
70c354e4abc9804f1746c08ca18217c32905e462e36ce3be39e772c180e86039b2
783a2ec07a28fb5c55df06f4c52c9de2bcbf6955817183995497cea956ae515d22
61898fa051015728e5a8aaca68ffffffffffffff

Generator (g): 0x2

Sender public key:

0xb780d255b507b00dffe23836ac0abe029a9d85a2e2de770fa1245d5140493d0
7356051087dfe6169085946ba1c79eda8c66a619ele60cf87b2b93d5ee5a03c08d
16c026092151ae038586803e4cc8965bbb23d1d16b7b3a09d255e28b0df45dd26
bd0a55e5c58a0b11ab6546cfefb2d7d8d8313ac92b09c5b5d32686d8532da53cf
dffebe54314ef96b11f5b9bd29471efba4afebd71b6d30b424e73bede0232b2794ca
6232ff9b5a693c3cbbc0ac368e75e914857c4a52e61fd836a4c718b25e01a8f8827
125222a8899a8c108fd49ca4e6f7fd5a60c933f4126e382da650373c92f45762b9a
de1fa50382eb8c52e6efd2898f827809ba9206f42854b77d5781

Connected to receiver

INSIDE KEY EXCHANGE (peer public key):

0x5d531fec6977f9422763984b54b7e14f9279d835db9bad4ef08475da53877c6a
2909c0a44b15bf8440e5bc73822e207727ce5a83dc172823638f24078a74752b85
2c105b19278b3cc382dd00d5fa807e728ec34cb7c077e92cf875f52f3f816d92e41c
92e83d28c90b722dlac69f81828630a3dlfb91e4f312eb8c4632960f7398ad39d03
5675a53eb0ec73c2bd9cddf68cdcd5d78d95c1da6866314c2d40594f0c3923b56
f7ab07f2e82a5bc8cbac33f707485b10e789f0f04306191798d8754f392dl349c24c
d7b77a3ae18c8eab7827edec101c51c475886fd48ba2682c2af0f2ab825409dbb8
da78ad7ae9e28dl84l38539lf45dccb7c523a8c7dd6e6744

Shared key length: 256 bytes

Shared key:

b'\xbb|\x08Z\x87\x8e\x92\xe1\xe3\xbf\xb9O\xb6l\xbc\xcc;\xf9\x04M\xc2\x
f3=!uj>\xa8\x04\xe7\x00L\xad\x86\xeb4\xfd\xc7\x82U\xe5\x0c\xb9\x10\x
b7\x84\x0f{\xcb\xe9_\xc5\xd8\xba\x0e\x18\xe0\xbd\xe3\xf7\x00\xa1\x
f0\xd7\x88\xcc0\x81\xc0\xb8\xe6\x9f\x06\xe0\xc4\xc7\xa9\x8de\x17\x95
\xc0NZ\$\xcceVK\x04\xf1\x1b&\xfb@V5@TB\x06\x15|\|. \x12\x15R\xf3\xa7V\x
a5&\xa8|vIR\xf4\xcd\xa1\xc2}\x92\xac\$\xa6\x88\x87\x16\xa8\x18\x16}^e\x
81\xb4\xed\xfdc\xdd\xef\x02\xe5z\xa3)\xfe\x82\x80\xbe\xe1oe\x8b\xcf\x
x1l\x0c|r\xb7F^9\x8b\x00\xc8\x12\x8e\xff8\x81A\xf7\x8f>\xc76\xa0@\x17>~
\xba\xb6\xa9\x1l\x8f\xf9
,\xea\x0b|(\xa8\xb0\xb97\x1b\xe7\x8c\xe6z7\ '@N|r\xc8\ 'xbd\xc6\xaf\x0
1C\xef\x95\xb0ga\xd3\xea\x0f\x97WX\x80\xa9\x0bq"S\xfd\xb8#\xe4\xee
\xef\xa0\x81\xe8\xf4\x04\xde\xa2_\xec8*7\xbf\x83\xa8\xd4\x07q'

INSIDE SEED ENCRYPTOR (key):

b'\xfe\x10"\xe3n78y%\xd57\xe6\x0e\xc6\x9e\xb0\xcb\xd8\xb75\x126\xa6;
\xc5H4\xb6Wx\xe3\xf4' of length 32 bytes

INSIDE SEED ENCRYPTOR (seed): 116197241997750167055934171051957910961

Encrypted seed:

b"\xc7\$]]*\xf4\xbb\xa7\xf1\xe7%\xeb\x98\xcd\x9a\x9e\x85\xfc\x8aFj\xc8
\xad\xf3|\xae\xaf\xf0'\xb8A\xba|\x96u\xa6<\x94>\x08\x08\xa0\xcd*\xcb
\x88\x8dR"

Encrypted seed length: 48 bytes

INSIDE HMAC:

b'\xa5\x8b\x87@\xdd\x8c\x15\xb45\x9b\xdl\xe0\xa0\x8d\xb2\xc6%\ \Dy?
\xdd C\xcf5T0\xf0\xbf\x87' of length 32 bytes

INSIDE STREAM CIPHER:

b'!@#\$%^&*()_+-=[]{|:; ",.<>?/~\n \xc2\xa1\xe2\x84\xa2\xc2\xa3\xc2\xa2
\xe2\x88\x9e\xc2\xa7\xc2\xb6\xe2\x80\xa2\xc2\xaa\xc2\xba\xe2\x80\x
93\xe2\x89\xa0\xc5\x93\xe2\x88\x91\xc2\xb4\xc2\xae\xe2\x80\xa0\xc2
\xa5\xc2\xa8\xcb\x86\xc3\xb8\xcf\x80"\ ' \xc2\xab\xe2\x89\xa4\xe2\x89
\xa5\xc3\xb7\n \xf0\x9f\x8c\x9f \xf0\x9f\x91\x8b \xf0\x9f\x94\x92

\xf0\x9f\x97\x9d\xef\xb8\x8f \xf0\x9f\x92\xbb \xf0\x9f\x93\xa1
\xf0\x9f\x8c\x90 \xe2\x9a\xa1 \xf0\x9f\x8e\xaf
\xf0\x9f\x8e\xb2\n\xd7\x9e\xd7\xa8\xd7\x97\xd7\x91\xd7\x99\xd7\x9d
\xd7\xa2\xd7\x91\xd7\xa8\xd7\x99\xd7\x99\xd7\x9d
\xd9\x88\xd8\xa7\xd9\x84\xd8\xb9\xd8\xb1\xd8\xa8\xd9\x8a\xd8\xa9
\xe1\x80\xa1\xe1\x80\x80\xe1\x80\xb9\xe1\x80\x81\xe1\x80\x9b\xe1\x80\xac'

Cipher text:

b';Dcd\xfc\xe0e\x98t\xeb\xa3\xd2\xdf{Tm\xd4<d\x0b\xab\xcf\xdcA\xcc\x0epjB\xef\x9b(q)r@\x7f\xc3\xac\xcb\x843+0\x17\xc2-\x95\xccD/'\x13\xbb\x7uX\xa2\xb6e\x8946\xd3Q\xd2#)\xf4Q'\x0f4\xe5,\xb2}\xfc3\x95\x1c\xabr\x1f\xc2\xaaam\r\xac\x89\x0f\x9buo\xc0\xbc\xc9\x94\x8c\xd2\xab|\xb6:\xc15Z\xa0`\xedo\xaeU\x045x6\xd4{\x93\xac\x84l\x00\xd6?#\xe9\xe6Z}| \x93\xec\x12\x93\xcce\x94\x0c\xb7g\xc9\x95\xb9\x98\xce\x8b\x13\xfd\xcl\xe0\xc6|\xc8\xff\xefb&\xd1\xd3Y\xb6u\x01\$\xb4\xdaQ\x8dw&\xe6\x162\xa5\x9b\x11\x8e\xaf\xcl\xb8L\x85\xa7J\xc5H\x14Vi\x95\xd1\xda\x4\xfc\xb9\xdc@ore\xae"\xdea\x95\x02' of length 209 bytes

Sent chunk 1/21

Sent chunk 2/21

Sent chunk 3/21

Sent chunk 4/21

Sent chunk 5/21

Sent chunk 6/21

Sent chunk 7/21

Sent chunk 8/21

Sent chunk 9/21

Sent chunk 10/21

Sent chunk 11/21

Sent chunk 12/21

Sent chunk 13/21

Sent chunk 14/21

Sent chunk 15/21

Sent chunk 16/21

Sent chunk 17/21

Sent chunk 18/21

Sent chunk 19/21

Sent chunk 20/21

Sent chunk 21/21

All chunks sent

Receiver Side:

Receiver Public Key Components:

Prime modulus (p):

0xffffffffffffc90fdaa22168c234c4c6628b80dc1cd129024e088a67cc74020bbea
63b139b22514a08798e3404ddef9519b3cd3a431b302b0a6df25f14374fe1356d6d5
1c245e485b576625e7ec6f44c42e9a637ed6b0bff5cb6f406b7edee386bfb5a899
fa5ae9f24117c4b1fe649286651ece45b3dc2007cb8a163bf0598da48361c55d39a6
9163fa8fd24cf5f83655d23dca3ad961c62f356208552bb9ed529077096966d670c
354e4abc9804f1746c08ca18217c32905e462e36ce3be39e772c180e86039b2783a
2ec07a28fb5c55df06f4c52c9de2bcbf6955817183995497cea956ae515d2261898f
a051015728e5a8aaca68ffffffffffff

Length of prime modulus (p): 256 bytes

Generator (g): 0x2

Receiver public key:

0x5d531fec6977f9422763984b54b7e14f9279d835db9bad4ef08475da53877c6a2
909c0a44b15bf8440e5bc73822e207727ce5a83dc172823638f24078a74752b852
c105b19278b3cc382dd00d5fa807e728ec34cb7c077e92cf875f52f3f816d92e41c92
e83d28c90b722dlac69f81828630a3dlfb91e4f312eb8c4632960f7398ad39d03567
5a53eb0ec73c2bd9cddf68cdcd5d78d95clda6866314c2d40594f0c3923b56f7a
b07f2e82a5bc8cbac33f707485b10e789f0f04306191798d8754f392d1349c24cd7b
77a3ae18c8eab7827edec101c51c475886fd48ba2682c2af0f2ab825409dbb8da78
ad7ae9e28d1841385391f45dccf7c523a8c7dd6e6744

Waiting for connection...

Connected by ('127.0.0.1', 48172)

INSIDE KEY EXCHANGE (peer public key):

0xb780d255b507b00dffe23836ac0abe029a9d85a2e2de770fa1245d5140493d07
356051087dfe6169085946ba1c79eda8c66a619ele60cf87b2b93d5ee5a03c08d16
c026092151ae038586803e4cc8965bbb23d1d16b7b3a09d255e28b0df45dd26bd

0a55e5c58a0b11ab6546cfef2d7d8d8313ac92b09c5b5d32686d8532da53cfdffe
be54314ef96b11f5b9bd29471efba4afebd71b6d30b424e73bede0232b2794ca623
2ff9b5a693c3cbbc0ac368e75e914857c4a52e61fd836a4c718b25e01a8f88271252
22a8899a8c108fd49ca4e6f7fd5a60c933f4126e382da650373c92f45762b9ade1fa
50382eb8c52e6efd2898f827809ba9206f42854b77d5781

Shared key length: 256 bytes

Shared key:

b'\xbb|\x08Z\x87\x8e\x92\xe1\xe3\xbf\xb9O\xb6l\xbc\xcc;\xf9\x04M\xc2\
xf3=!uj>\xa8\x04\xe7\x00L\xad\x86\xeb4\xfd\xc7\x82U\xe5\x0c\xb9\x10\x
b7\x84\x0f{\xcb\xe9_\xc5\xd8\xba\x0e\x18\xe0\xbd\xe3\xf7\x00\xa1\xf0
\xd7\x88\xcc0\x81\xc0\xb8\xe6\x9f\x06\xe0\xc4\xc7\xa9\x8de\x17\x95\x
c0NZ\$\xcceVK\x04\xf1\x1b&\xfb@V5@TB\x06\x15\|. \x12\x15R\xf3\xa7V\xa5
&\xa8vIR\xf4\xcd\xa1\xc2}\x92\xac\$\xa6\x88\x87\x16\xa8\x18\x16}^e\x81
\xb4\xed\xfdc\xdd\xef\x02\xe5z\xa3)\xfe\x82\x80\xbe\xe1oe\x8b\xcf\x11\
x0c\r\xb7F^9\x8b\x00\xc8\x12\x8e\xff8\x81A\xf7\x8f>\xc76\xa0@\x17>~\xb
a\xb6\xa9\x11\x8f\xf9
,\xea\x0b|(\xa8\xb0\xb97\x1b\xe7\x8c\xe6z7\ '@N\r\xc8\ '\xbd\xc6\xaf\x01
C\xef\x95\xb0ga\xd3\xea\x0f\x97WX\x80\xa9\x0bq"S\xfd\xb8#\xe4\xee\
xef\xa0\x81\xe8\xf4\x04\xde\xa2_\xec8*7\xbf\x83\xa8\xd4\x07q'

INSIDE HMAC:

b'\xa5\x8b\x87@\xdd\x8c\x15\xb45\x9b\xd1\xe0\xa0\x8d\xb2\xc6%\ \Dy?
\xdd C\xcf5T0\xf0\xbf\x87' of length 32 bytes

INSIDE SEED ENCRYPTOR (key):

b'\xfe\x10"\xe3n78y%\xd57\xe6\x0e\xc6\x9e\xb0\xcb\xd8\xb75\x126\xa6;\
xc5H4\xb6Wx\xe3\xf4' of length 32 bytes

Decrypted seed: 116197241997750167055934171051957910961

INSIDE STREAM CIPHER: b';Dcd\xfc\xe0e\x98t\xeb'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xa3\xd2\xdf{Tm\xd4<d\x0b'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xab\xcf\xdcA\xcc\x0epjB\xef'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\x9b(q)r@\x7f\xc3\xac\xcb'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\x843+0\x17\xc2-\x95\xccD'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'"/\x13\xbb\x7uX\xa2\xb6e"

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\x8946\xd3Q\xd2#)\xf4Q'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b""\x0f4\xe5,\xb2}\xfc3\x95"

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\x1c\xabr\xlf\xc2\xaam\r\xac\x89'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\x0f\x9buo\xc0\xbc\xc9\x94\x8c\xd2'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b"\xab|\xb6:\xc15Z\xa0~"

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\xedo\xaeU\x045x6\xd4{'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\x93\xac\x84l\x00\xd6?#\xe9\xe6'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'Z}|\x93\xec\x12\x93\xce\x94'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\x0c\xb7g\xc9\x95\xb9\x98\xce\x8b\x13'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\xfd\xcl\xe0\xc6|\xc8\xff\xefb&'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\xd1\xd3Y\xb6u\x01\$\xb4\xdaQ'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\x8dw&\xe6\x162\xa5\x9b\x11\x8e'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\xaf\xcl\xb8L\x85\xa7J\xc5H\x14'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'Vi\x95\xd1\xda\xc4\xfc\xb9\xdc@'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'ore\xae"\xdea\x95\x02'

Received chunk, decrypted 9 bytes
Decryption complete. Saved to decrypted.txt

Test 3:

Plain Text:

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor
incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis
nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.
Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore
eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt
in culpa qui officia deserunt mollit anim id est laborum.

Multiple paragraphs to test chunking and buffering.
Testing long message transmission and chunk handling.
Verifying stream cipher operation on large data.
Testing memory usage and performance.

Sender Side:

Message to send: b'Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor \nincididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis \nnostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.\nDuis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore \neu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt\nin culpa qui officia deserunt mollit anim id est laborum.\n\nMultiple paragraphs to test chunking and buffering.\nTesting long message transmission and chunk handling.\nVerifying stream cipher operation on large data.\nTesting memory usage and performance.'

Random seed: 289916329629684850162300570350976969360

Sender Public Key Components:

Prime modulus (p):

0xffffffffffffffffc90fdaa22168c234c4c6628b80dc1cd129024e088a67cc74020bbea63b139b22514a08798e3404ddef9519b3cd3a431b302b0a6df25f14374fe1356d6d51c245e485b576625e7ec6f44c42e9a637ed6b0bff5cb6f406b7edee386bfb5a899fa5ae9f24117c4b1fe649286651ecee45b3dc2007cb8a163bf0598da48361c55d39a69163fa8fd24cf5f83655d23dca3ad961c62f356208552bb9ed529077096966d670c354e4abc9804f1746c08ca18217c32905e462e36ce3be39e772c180e86039b2783a2ec07a28fb5c55df06f4c52c9de2bcbf6955817183995497cea956ae515d2261898fa051015728e5a8aacaa68ffffffffffffff

Generator (g): 0x2

Sender public key:

0x25ff6c69a9f3cc9e3af0b444bb678716f5925aa53f805e71a7efee6501ea957483b6bf326afa6b9901d49fbd65ebe2fb3b052b73be0aa188fbc3285bc4a35f7ab0478479a10f53a6fd47ec112a9cdd61eeae0335b89ba44fd5ec66e7dcc01555ac8cdea48021f480c7bd852070114e9a36482acc63da321c0d2d9129757ec1afa34fb3da1c1350e083e27ca35aee7bed9bff5f9df2cf842af4224ad213accfbea46f20de009a4caf8ca5ca3a5a603c7b592d9b44a660a5e9d344a9c1e2391238d6906e08dde21efdb074aa94e0c7168b49fc9c2118ee77d733aa52c63832f4e0528951cfd9de9f14805820a5f796245764f801ea17c89cc96ed519b3d4f01a71

Connected to receiver

INSIDE KEY EXCHANGE (peer public key):

0x590bfbe95b51ee095286e9ee222f2c78e36ea3dcfcc1c42a60429e53a92c270bf

530ee2d481a514d73574d70753535cfec601997685603873d59df3ba5ea910ae795
03f04d663bfb8626c51502af6184a075e08f1d47cabbbae763a5a4d8bdbb9cc0b7d
82ada11bd5715a95aa6f924631b2a48b0fc518bbb2942899497010c736781733f48a
c836529a427c7a3926d95523d959e3699f1ff0f6b3a0ecba88167f2839745750c2e
aeb2591b368d9886f65303c2ad4b384e04f6c4f580848c34f8eec09c6e1f302acf5
bafb6b7a5774c63301aa6ac40a9f5c083baa4c317ac64c9f8fc12b238069651bcd9
fc3ecb8d5b03438b0ce5fe2f7be1f8b9735b42466e88d

Shared key length: 256 bytes

Shared key:

b'\xed\x04\xbf\x1d\xc9\x93:)]=\x1a\xd6\xbea\xde=X\x7w\xeb\xfe]Gd\xd8
nIGuD\x0f8\xf3\x82\xcb\x82\xee\x00\x85\xbfC\x05&\x12\x02\xaa\xc6\xe8
\xca\xba\x90\xfa\xag\xal\xda\xbb\x7v\x1b/\xc5\x17H;\x10\x87\x10+\xf8\xf1
\xc7M\xbf\xa9\xb4\xd9Y\xe78\x16,n\x00\xb6\x85\xeaO\xcb:D\x1a\x8c"6q#\xa9_
\xaeLp\xdb\x85n6O\x87U\xelm\xc2\tG\xd7\|;7x\xf7\x1f\xe5v\xd6Q\x9c\xc
3\x8e\xd2`\xcf\xf1>\xf9\xd1\xef\x8e\x05\xdf\xb8\xca\x9bC?\xe8\xb0/\xf8
\x1e\x98\x15\|\xec\xd3\x98q\x98R\x9eTt\$\x80\x8a\x1d'\xdd\xe5\xe6?:(u14
\xde\xa7\x8c\xba\xa2\|B\xc0\$z\r0\xc5\xe2\x0e\xdc;\xc3\x94\xa1^|\xc2\
\x94\xc1\xb1#cqC\x99\xda\xe4\xb0\xf5\xd04\xad\xcf\xc2\x8b\nn\x98\x03
\xa1\xd1\x1e\x03CU\xdd[\xb4\x02X~\xa3\xb3\xd2\xc0:\xeb\x920\x04P'\x
03dJ\x11\xda\x04\xf5\x1c\x00!\xa8\xa6'

INSIDE SEED ENCRYPTOR (key):

b'IU\xaa\x92bUh#\x81|bt\xae\x83\r\x04\xbf\xef\xe6!"<\xed\x80\xe5}\xd1>\xd6:n_' of length 32 bytes

INSIDE SEED ENCRYPTOR (seed): 289916329629684850162300570350976969360

Encrypted seed:

b'\xf9@\xb0\r\x10\x8e\xb0;\xd5\xb4\xb1r\x80l\x9b\xf6\xa4\xd0\n2^\xa7\xcd\xbbH:\xa8\xd4\xde\xb0\xee\xdeF\xd0BR\x16\xca3\xd6\xc4M\xd9F\xf9\xfc\xe1\xc2'

Encrypted seed length: 48 bytes

INSIDE HMAC:

b'\x83YQ\x97IK+\x16\x81^J\xe6\xb1Z\xa6\xb0\xa9\xc7-\x9b\xa2/\x17~\x9fn

5\xef9\x8a\xa6\xb2' of length 32 bytes

INSIDE STREAM CIPHER: b'Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor \nincididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis \nnostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.\nDuis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore \neu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt\nin culpa qui officia deserunt mollit anim id est laborum.\n\nMultiple paragraphs to test chunking and buffering.\nTesting long message transmission and chunk handling.\nVerifying stream cipher operation on large data.\nTesting memory usage and performance.'

Cipher text:

b"X\xa3\xb30\x85\xef\x1c?\xd9L\x87\$W\xe6;t\xb9*<\xed\x5t)rn?\x03Q\x1de\xe2>\xea1\x98\xbf~\xc3\x84\xf5w\xb8W\xfb-\xc3sw\x07\xea\xbd\x18%j\xad\x7f\x0c\x07\xf4\xb3\x81\x10\xb2\xc5\$ou\t\x7c\x9e!u9\xf5\x0c\x1lg\x17\x82SdD\xae\xacu\xad\xc3\x85\xc5\x95\xf2\x03\xbb\xffuF\xbb!\x8bW\xda\x07\x89X/r?8\x9ly\xbbp\x91\xd8<o\xb2}M\x8e\x00\x15\x96L\xab\x98j\xcd^\x17&\xe9\x04\|\x90P#X\x10\x16\xa1@\xd3\x0f\x87y\xef\xc9\xdc?\xe7\xb5\xdc\x1b\xbf<\xd8f\x0b\xc3\xce\x1a\xa7\xa5\x94S\x19\xc3\x14\xb9\xfb5\x07%5M\xbd\xfe\x89p\xdb\xebW\x85m\xa7\xb2\x06\x05\xf6\xb1\xcaT\x05*\x00\xde\xd2\xacn+\xc3:\xd8\x06\xf9\xb1_p>p\x8c\xaf\xc2\x80F\xb4\xe0\xa36\xf5\x80\xe6\xb3\xeb!\x16\x01qL\xff\x3e)\xe9d\x8fa\xe3\xldK\xb9\xc9\xf8\x89\xcds\x9b\x92\xc5\xd0\xe4&\x88c\x01\x96\xf0B\x9f\nh\xlfW\x7f6>q\xcdJ\|\|\x1b\xbbH}\xc9\xc0\x12h\xbf\x9lg\xe2\xcf\x94B6\xea\xdb!\xbb3\xd5\xbd\$)\x08\x8a\xf9\$X\xa0\xec\xdf\x3\xfdn\xcdX5\x1e\xdb\x03\xc8\|\|\xd2\x8c\xd2L\xea\x81l\x814\x8d\x10\xab\x87\xa2\x0e\x1aG\xea\x82\x81:/\xf9\xd9\x07\xdb{\xf8J\xc5\xbc\x1d>Sy\x9a\x8c\xe3\x19\$\x17p\xde^\xf3\xde\xc7\x05\x18\x1a\xa1\xb2\x9e\xea\xa5\xec\x9c\xa3=\xf7\xca\xc7}\x81\t\xa3\xca\xedx\xa1*\xc1\xfaB\x8d\|\|\x1ar\x043*9\xa6\xaf\xa9pQ\xe7\x01r\xb5.\ro\xb6\x8e@\xa4+\xb0!\x1e\x19M\xfbw\xb4\x9e\x0e\xdb\x98\x0fveuE\x04\x0c\x08?g\xfcA\x1a\xa2\xd8\xc5\xf6\x87\x0e\x8dTW\xf1\x9dG\xc9\x08\x0e\x89o\xfe|\xca\xc2<kbs\xfaY\x0cA\x97\x0b\x84\xd8\xd8\xe0\xb4\x19e\x9dn\x86\xc7\x99\x13\xa9\x9a'm\x07g&b`Ny\x02\xa9\x97H\xea\xdb\xfe/L!* \xa5\xb5\xd78\xean\xa6\x0fk\x99\xe2E\xd2z\nZ\x1b\x0e\$\x90\xfan\xa2\x84\x08\xde\|\|\x92\xcdF.\x8b\x13\xf3\xdc\x08\x91\xb0\xf9<\x7f1\xa47\x04\xe4\x1a\x8c\x04\xe7;Rz\xei\x89\rN\xd3|ic\xdd\x99*#/[\xfc\x8e\x95\xe0\x97j\xab\x86\xe7za\xcbc\xci\xd6\x80\x1a^S\xb8\xb1\xfb\x87\x82\xe7m\xff\r\xdey\x96\xa9\xf9\x15\x

xe6\xe7\xc5\xb7hD\x12\x7f\xe1\xq9\xdd\xd7\n\xf2HT\xd1\x90\x96\x03\xfc
\x01\x91\xc4\xf0\xceL\xe6\x80\xeb\\\xd8T3" of length 642 bytes

Sent chunk 1/65

Sent chunk 2/65

Sent chunk 3/65

Sent chunk 4/65

Sent chunk 5/65

Sent chunk 6/65

Sent chunk 7/65

Sent chunk 8/65

Sent chunk 9/65

Sent chunk 10/65

Sent chunk 11/65

Sent chunk 12/65

Sent chunk 13/65

Sent chunk 14/65

Sent chunk 15/65

Sent chunk 16/65

Sent chunk 17/65

Sent chunk 18/65

Sent chunk 19/65

Sent chunk 20/65

Sent chunk 21/65

Sent chunk 22/65

Sent chunk 23/65

Sent chunk 24/65

Sent chunk 25/65

Sent chunk 26/65

Sent chunk 27/65

Sent chunk 28/65

Sent chunk 29/65

Sent chunk 30/65

Sent chunk 31/65

Sent chunk 32/65

Sent chunk 33/65

Sent chunk 34/65

Sent chunk 35/65

Sent chunk 36/65

Sent chunk 37/65
Sent chunk 38/65
Sent chunk 39/65
Sent chunk 40/65
Sent chunk 41/65
Sent chunk 42/65
Sent chunk 43/65
Sent chunk 44/65
Sent chunk 45/65
Sent chunk 46/65
Sent chunk 47/65
Sent chunk 48/65
Sent chunk 49/65
Sent chunk 50/65
Sent chunk 51/65
Sent chunk 52/65
Sent chunk 53/65
Sent chunk 54/65
Sent chunk 55/65
Sent chunk 56/65
Sent chunk 57/65
Sent chunk 58/65
Sent chunk 59/65
Sent chunk 60/65
Sent chunk 61/65
Sent chunk 62/65
Sent chunk 63/65
Sent chunk 64/65
Sent chunk 65/65
All chunks sent

Receiver Side:

Receiver Public Key Components:

Prime modulus (p):

0xffffffffffffffffc90fdaa22168c234c4c6628b80dc1cd129024e088a67cc74020bbe
a63b139b22514a08798e3404ddef9519b3cd3a431b302b0a6df25f14374fe1356d6

d51c245e485b576625e7ec6f44c42e9a637ed6b0bff5cb6f406b7edee386bfb5a
899fa5ae9f24117c4b1fe649286651ece45b3dc2007cb8a163bf0598da48361c55d
39a69163fa8fd24cf5f83655d23dca3ad961c62f356208552bb9ed52907709696
6d670c354e4abc9804f1746c08ca18217c32905e462e36ce3be39e772c180e860
39b2783a2ec07a28fb5c55df06f4c52c9de2bcbf6955817183995497cea956ae51
5d2261898fa051015728e5a8aaca68ffffffffffff

Length of prime modulus (p): 256 bytes

Generator (g): 0x2

Receiver public key:

0x590bfbe95b51ee095286e9ee222f2c78e36ea3dcfcc1c42a60429e53a92c270
bf530ee2d481a514d73574d70753535cfec601997685603873d59df3ba5ea910ae
79503f04d663bfb8626c51502af6184a075e08fld47cabbbae763a5a4d8bdbb9cc
0b7d82ada11bd5715a95aa6f924631b2a48b0fc518bbb2942899497010c7367817
33f48ac836529a427c7a3926d95523d959e3699f1ff0f6b3a0ecba88167f283974
5750c2eaeb2591b368d9886f65303c2ad4b384e04f6c4f580848c34f8eec09c6
elf302acf5bafb6b7a5774c63301aa6ac40a9f5c083baa4c317ac64c9f8fc12b238
069651bcd9fc3ecb8d5b03438b0ce5fe2f7be1f8b9735b42466e88d

Waiting for connection...

Connected by ('127.0.0.1', 37014)

INSIDE KEY EXCHANGE (peer public key):

0x25ff6c69a9f3cc9e3af0b444bb678716f5925aa53f805e71a7efee6501ea95748
3b6bf326afa6b9901d49fbd65ebe2fb3b052b73be0aa188fbc3285bc4a35f7ab0
478479a10f53a6fd47ec112a9cdd61eeae0335b89ba44fd5ec66e7dcc01555ac8c
dea48021f480c7bd852070114e9a36482acc63da321c0d2d9129757ec1afa34fb3
dalc1350e083e27ca35aee7bed9bff5f9df2cf842af4224ad213accfbea46f20de0
09a4caf8ca5ca3a5a603c7b592d9b44a660a5e9d344a9cle2391238d6906e08
dde21efdb074aa94e0c7168b49fc9c2118ee77d733aa52c63832f4e0528951cfd9d
e9f14805820a5f796245764f801ea17c89cc96ed519b3d4f01a71

Shared key length: 256 bytes

Shared key:

b'\xed\x04\xbf\x1d\xc9\x93:)]=\x1a\xd6\xbea\xde=X\x7w\xeb\xfe]Gd\x
d8nIGuD\x0f8\xf3\x82\xcb\x82\xee\x00\x85\xbfC\x05&\x12\x02\xaa\x6
\xe8\xca\xba\x90\xfa\xga\xda\xb3\xd7v\x1b/\xc5\x17H;\x10\x87\x10+\x

8f\xf1\xc7M\xbf\xa9\xb4\xd9Y\xe78\x16,n\x00\xb6\x85\xeaO\xcb:D\x1a\x8c"6q#\xa9_
\xaeLp\xdb\x85n6O\x87U\xelm\xc2\tG\xd7\ ;7x\xf7\x1f\xe5v\xd6Q\x9c\x3\x8e\xd2`\xcf\xf1>\xf9\xd1\ref\x8e\x05\xdf\xb8\xca\x9bC?\xe8\xb0/\x8f\x1e\x98\x15\ |\xec\xd3\x98q\x98R\x9eTt\$\x80\x8a\x1d\ '\xdd\xe5\xe6?:(u14\xde\xa7\x8c\xba\xa2\ |B\xc0\$z\r0\xc5\xe2\x0e\xdc;\xc3\x94\xa1^|\xc2\x94\xcl\xbl#cqC\x99\xda\xe4\xb0\xf5\xd04\xad\xcf\xc2\x8b\nn\x98\x03\xa1\xd1\x1e\x03CU\xdd[\xb4\x02X~\xa3\xb3\xd22\xc0:\xeb\x920\x04P\ '\x03dJ\x1l\xda\x04\xf5\x1c\x00!\xa8\xa6'

INSIDE HMAC:

b'\x83YQ\x97IK+\x16\x81^J\xe6\xb1Z\xa6\xb0\xa9\xc7-\x9b\xa2/\x17~\x9fn5\xef9\x8a\xa6\xb2' of length 32 bytes

INSIDE SEED ENCRYPTOR (key):

b'IU\xaa\x92bUh#\x81|bt\xae\x83\r\x04\xbf\xef\xe6l"<\xed\x80\xe5}\xd1>\xd6:n_' of length 32 bytes

Decrypted seed: 289916329629684850162300570350976969360

INSIDE STREAM CIPHER: b'X\xa3\xb30\x85\xef\x1c?\xd9L'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\x87\$W\xe6;t\xb9*<\xed'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xd5t)rn?\x03Q\x1de'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xe2>\xea1\x98\xbf~\xc3\x84\xf5'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'w\xb8W\xfb-\xc3sw\x07\xea'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xbd\x18%j\xad\x7f\x0c\x07\xf4\xb3'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\x81\x10\xb2\xc5\$ou\t\xc7\x9e'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'!u9\x05\x0c\x11g\x17\x82S'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'dD\xae\xacu\xad\xc3\x85\xc5\x95'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\xf2\x03\xbb\xffuF\xbb!\x8bW'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\xda\x07\x89X/r?8\x91y'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\xbbp\x91\xd8<o\xb2}M\x8e'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\x00\x15\x96L\xab\x98j\xcd^\x17'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'&\xe9\x04\\\x90P#X\x10\x16'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\xa1@\xd3\x0f\x87y\xef\xc9\xdc?'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\xe7\xb5\xdc\x1b\xbf<\xd8f\x0b\xc3'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\xce\x1a\xa7\xa5\x94S\x19\xc3\x114'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\xb9\xfb5\x07%5M\xbd\xfe\x89'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'p\xdb\xebW\x85m\xa7\xb2\x06\x05'

Received chunk, decrypted 10 bytes
INSIDE STREAM CIPHER: b'\xf6\xb1\xcaT\x05*\x00\xde\xd2\xac'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'n+ \xc3:\xd8\x06\xf9\xb1_p'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'>p\x8c\xaf\xc2\x80F\xb4\xe0\xa3'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'6\xf5\x80\xe6\xb3\xeb!\x16\x01q'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'L\xff\xe3)\xe9d\x8fa\xe3'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\x1dK\xb9\xc9\xf8\x89\xcdS\x9b\x92'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xc5\xd0\xe4&\x88c\x01\x96\xf0B'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\x9f\nh\x1fW\x7f6>q\xcd'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'J \ \x1b\xbbH}\xc9\xc0\x12h'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xbf\x91g\xe2\xcf\x94B6\xea\xdb'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'!\xbb3\xd5\xbd\$)\x08\x8a\xf9'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\$X\xa0\xec\xdf\xd3\xfdn\xcdX'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'5\x1e\xdb\x03\xc8 \ \xd2\x8c\xd2L'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xea\x81I\x814\x8d\x10\xab\x87\xa2'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\x0e\x1aG\xea\x82\x81:/\xf9\xd9'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\x07\xdb{\xf8J\xc5\xbc\x1d>S'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'y\x9a\x8c\xe3\x19\$\x17p\xde^'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xf3\xde\xc7\x05\x18\x1a\xa1\xb2\x9e\xea'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xa5\xec\x9c\xa3=\xf7\xca\xc7}\x81'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\t\xa3\xca\xedx\xa1*\xc1\xfaB'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\x8d\ \x1ar\x043*9\xa6\xaf'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xa9pQ\xe7\x01r\xb5.\ro'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xb6\x8e@\xa4+\xb0!\x1e\x19M'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xfbw\xb4\x9e\x0e\xdb\x98\x0fve'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'uE\x04\x0c\x08?g\xfcA\x1a'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xa2\xd8\xc5\xf6\x87\x0e\x8dTW\xf1'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\x9dG\xc9\x08\x0e\x89o\xfe|\xca'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xc2<kbs\xfaY\x0cA\x97'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\x0b\x84\xd8\xd8\xe0\xb4\x19e\x9dn'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b"\x86\xc7\x99\x13\xa9\x9a'm\x07g"

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'&b`Ny\x02\xa9\x97H\xea'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xdb\xfe/L!* \xa5\xb5\xd78'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xean\xa6 \x0fk\x99\xe2E\xd2'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'z\nZ\x1b\x0e\$\x90\xfan\xa2'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\x84\x08\xde \ \x92\xcdF.\x8b\x13'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xf3\xdc\x08\x91\xb0\xf9<\x7f1\xa4'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'7\x04\xe4\x1a\x8c\x04\xe7;Rz'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xe1\x89\rN\xd3|ic\xdd\x99'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'*#[/\xfc\x8e\x95\xe0\x97j'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xab\x86\xe7za\xcbc\xcl\xd6\x80'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\x1a^\xb8\b1\xfb\x87\x82\xe7m'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xff\r\xdey\x96\xa9\xf9\x15\xe6\xe7'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\xc5\xb7hD\x12\x7f\xe1\xa9\xdd\x7'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\n\x2HT\x1\x90\x96\x03\xfc\x01'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'\x91\xc4\xf0\xceL\xe6\x80\xeb\\ \xd8'

Received chunk, decrypted 10 bytes

INSIDE STREAM CIPHER: b'T3'

Received chunk, decrypted 2 bytes

Decryption complete. Saved to decrypted.txt